

VEDANT PUNGLIYA

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EDUCATION

Iowa State University

Ames, IA

Bachelor of Science, Computer Science

Expected May 2026

GPA: 3.99/4.00 — Minors: Cyber Security Engineering, Data Science

Coursework: Analysis of Algorithms, Machine Learning, Deep Learning, Operating Systems, Cryptography, DBMS

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL, R, Bash, JavaScript (Node.js), HTML/CSS

AI/ML: PyTorch, TensorFlow, LLMs, RAG, LangChain, Vector Databases (FAISS), Model Context Protocol (MCP)

Cloud/DevOps: AWS (EC2, ECS, Fargate, S3, IAM, VPC), Docker, Kubernetes, Jenkins, CI/CD, Terraform, Git

Backend/Web: Flask, FastAPI, Spring Boot, REST APIs, Microservices, Redis, RabbitMQ, PostgreSQL

PROFESSIONAL EXPERIENCE

PrecisePK

San Diego, CA

Software & Algorithm Engineering Intern

Sep 2025 – Dec 2025

- **Boosted convergence stability 15%** via **FOCE-I algorithms** in Python/C++ for nonlinear PK/PD modeling.
- **Cut runtime 70%** via scalable **AWS pipeline** (ECS, CodeBuild, S3, DynamoDB) for parameter estimation.
- **Raised clinical accuracy** by optimizing 1/2/3 compartment model functions for complex pharma datasets.

2GoProducts

San Diego, CA

Generative AI & Software Engineering Intern

Jun 2025 – Dec 2025

- **Built fault-tolerant multi-agent AI** (10K+ daily reqs, 99.9% uptime) via **MCP-based AWS** microservices.
- **Hit 85% NL-to-SQL accuracy** via **RAG** pipeline with **FAISS** vector search and **SQLGlot** schema parsing.
- **Cut analyst workload 60%** via multi-tenant GenAI on **ECS Fargate** with CI/CD for 11 microservices.

Iowa State University

Ames, IA

Undergraduate AI/ML Research Scientist

Aug 2024 – Jun 2025

- **Improved hallucination detection F1 by 40%** via latent-space analysis and saliency studies for Code LLMs.
- **Reduced insecure code generation 65%** by developing **activation steering guardrails** with LLM-as-judge.
- **Published AI safety findings** in peer-reviewed workshop after validating robustness via 200+ concept ablations.

Weekend AI

Pune, India

AI Engineering Intern

May 2024 – Aug 2024

- **Increased efficiency 90%** using **LangChain** and **CrewAI** multi-agent systems for autonomous data workflows.
- **Reduced ticket resolution time 40%** via an advanced **RAG** chatbot grounded in company knowledge bases.
- **Accelerated debugging 3x** via real-time observability using **Prometheus**, **Grafana**, and **Phoenix AI** tracing.

SELECTED PROJECTS

- **Nitro-Agent: Secure Enclave Orchestrator** — AI compiler converting Python scripts to AWS Nitro Enclaves. Automates vsock rewrites, Docker builds, and Terraform for attested enterprise-grade security.
- **BioSignify: ML E-Signature Platform** — Built a B2B platform using blockchain and vector databases; processes 100K+ authentications/day with 99.2% accuracy and sub-200ms latency.
- **Intelligent NLP-to-SQL Engine** — Developed a Transformer-based TEXT-to-SQL system optimizing 50K+ daily queries; democratized DB administration via prompt-based schema building.
- **TravelBuddy: AI Trip Planner** — Scaled personalized recommendations to 1000+ users at 99.99% uptime using Spring Boot, a hyper-personalized LLM engine, and automated CI/CD.

CERTIFICATIONS & HONORS

[AWS Certified Solutions Architect – Associate](#)

Nov 2025

Honors: Phi Beta Kappa, IEEE Eta Kappa Nu (HKN), Phi Kappa Phi, President's Excellence Scholarship

Leadership: CS Student Advisory Council; Workshop Lead – CS & Software Engineering Club